## Weak Entity Sets

Entity types that do not have key attributes of their own are called weak entity types.

In contrast, regular entity types that do have a key attribute are called strong entity types.

Entities belonging to a weak entity type are identified by being related to specific entities from another entity type in combination with one of their attribute values. We call this other entity type the identifying or owner entity type, and we call the relationship type that relates a weak entity type to its owner the identifying relationship of the weak entity type. A weak entity type always has a total participation constraint (existence dependency) with respect to its identifying relationship because a weak entity cannot be identified without an owner entity.

A weak entity type normally has a **partial key**, which is the attribute that can uniquely identify weak entities that are related to the same owner entity.

For example, suppose that employees can purchase insurance policies to cover their dependents. We wish to record information about policies, including who is covered by each policy, but this information is really our only interest in the dependents of an employee. If an employee quits, any policy owned by the employee is terminated and we want to delete all the relevant policy and dependent information from the database.

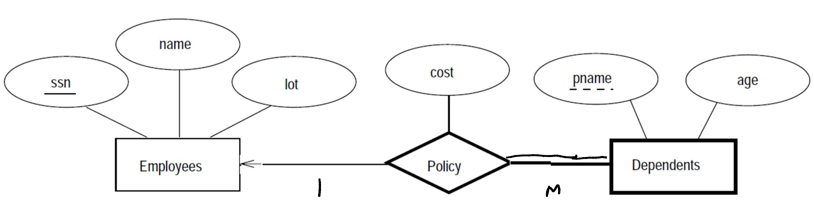
We might choose to identify a dependent by name alone in this situation, since it is reasonable to expect that the dependents of a given employee have different names. Thus the attributes of the Dependents entity set might be pname and age. The attribute pname does not identify a dependent uniquely.

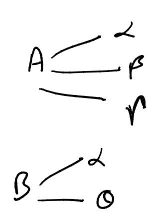
Dependents is an example of a **weak entity set**. A weak entity can be identified uniquely only by considering some of its attributes in conjunction with the primary key of another entity, which is called the **identifying owner**.

The following restrictions must hold:

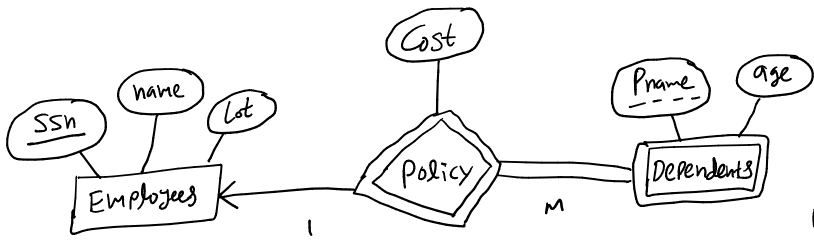
- The owner entity set and the weak entity set must participate in a **one-to-many relationship set** (one owner entity is associated with one or more weak entities, but each weak entity has a single owner). This relationship set is called the **identifying relationship set** of the weak entity set.
- The weak entity set must have **total participation** in the identifying relationship set.

For example, a Dependents entity can be identified uniquely only if we take the key of the owning Employees entity and the pname of the Dependents entity. The set of attributes of a weak entity set that uniquely identify a weak entity for a given owner entity is called a partial key of the weak entity set. In our example pname is a partial key for Dependents.




The total participation of Dependents in Policy is indicated by linking them with a dark/double line. The arrow indicates that each Dependents entity appears in at most one (indeed, exactly one, because of the participation constraint) Policy relationship. To underscore the fact that Dependents is a weak entity and Policy is its **identifying relationship**, we draw both with dark/double lines. To indicate that pname is a partial key for Dependents, we underline it using a broken line. This means that there may well be two dependents with the same pname value.




Emp(ssn, name, lot)

Dependents(ssn, Pname, age)

Policy(ssn, Pname, cost)

Emp(ssn, name, lot)

Dependent_Policy(ssn, Pname, age, cost)

In E R diagrams, a weak entity set is depicted via a double rectangle.

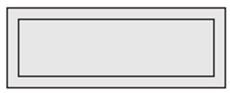
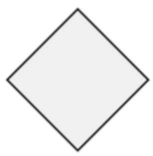
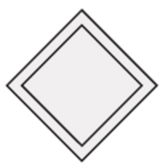
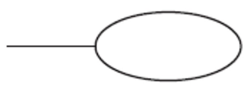
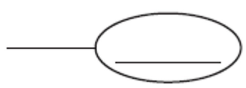
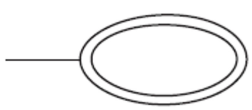
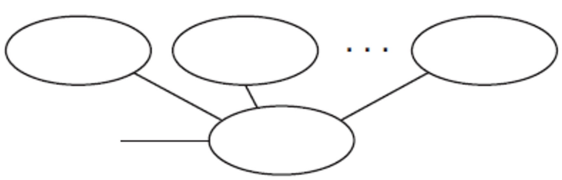
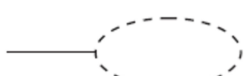
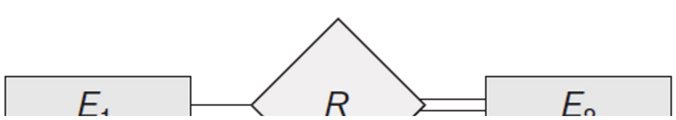
We underline the discriminator (partial key) of a weak entity set with a dashed line.

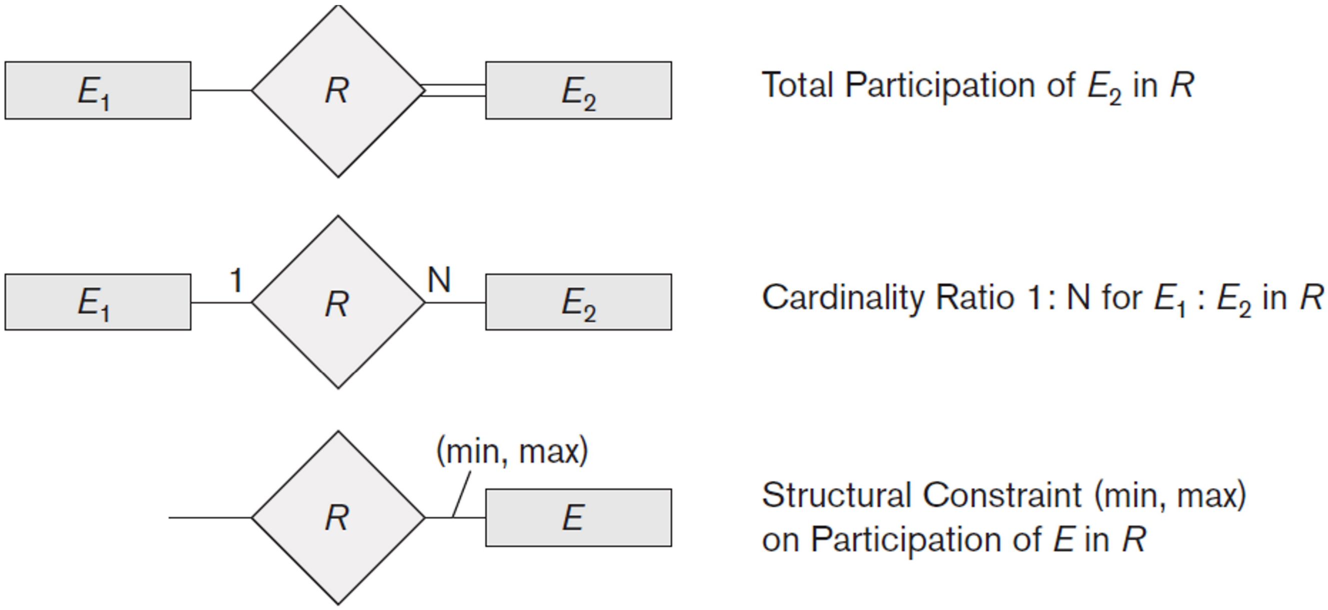
The relationship set connecting the weak entity set to the identifying strong entity set is depicted by a double diamond.

## Note:

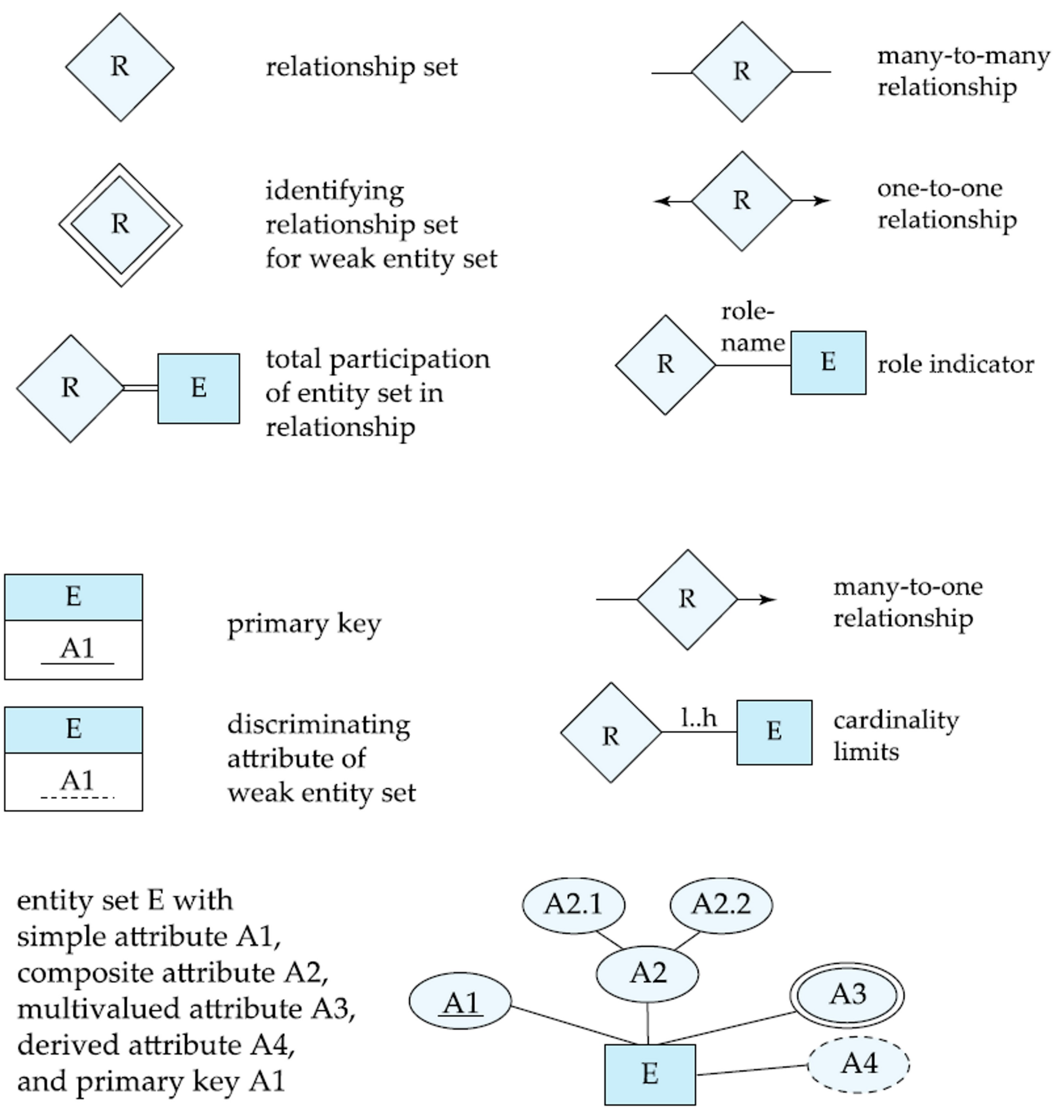
- The existence of a weak entity set depends on the existence of a **identifying entity set**.
- It must relate to the identifying entity set via a **total, one-to-many relationship set** from the identifying to the weak entity set.
- Identifying relationship depicted using a double diamond.
- The discriminator (or partial key) of a weak entity set is the set of attributes that distinguishes among all the entities of a weak entity set that depend on one particular strong entity.
- The primary key of a weak entity set is formed by the primary key of the strong entity set on which the weak entity set is existence dependent, plus the weak entity set's discriminator.

## Notation for ER diagrams:

| Symbol | Meaning |
| --- | --- |
|  | Entity |
|  | Weak Entity |
|  | Relationship |
|  | Indentifying Relationship |
|  | Attribute |
|  | Key Attribute |
|  | Multivalued Attribute |
|  | Composite Attribute |
|  | Derived Attribute |
|  | Total Participation of $E_2$ in $R$ |

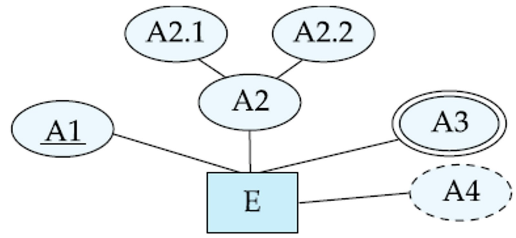


E1
R
E2
Total Participation of E2 in R
E1
1
R
N
E2
Cardinality Ratio 1: N for E1 : E2 in R
(min, max)
R
E
Structural Constraint (min, max)
on Participation of E in R

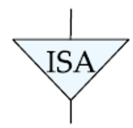
R
relationship set
R
many-to-many
relationship
R
identifying
relationship set
for weak entity set
R
one-to-one
relationship
R
E
total participation
of entity set in
relationship
R
role-
name
E
role indicator
E
A1
primary key
R
many-to-one
relationship
E
A1
discriminating
attribute of
weak entity set
R
l..h
E
cardinality
limits
entity set E with
simple attribute A1,
composite attribute A2,
multivalued attribute A3,
derived attribute A4,
and primary key A1
A2.1
A2.2
A2
A1
A3
A4
E

entity set E with
simple attribute A1,
composite attribute A2,
multivalued attribute A3,
derived attribute A4,
and primary key A1

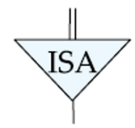



weak entity set

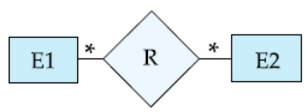
generalization



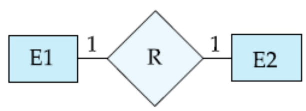

total
generalization



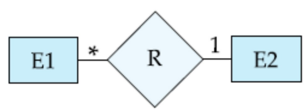

many-to-many
relationship



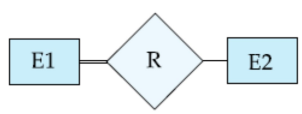

one-to-one
relationship



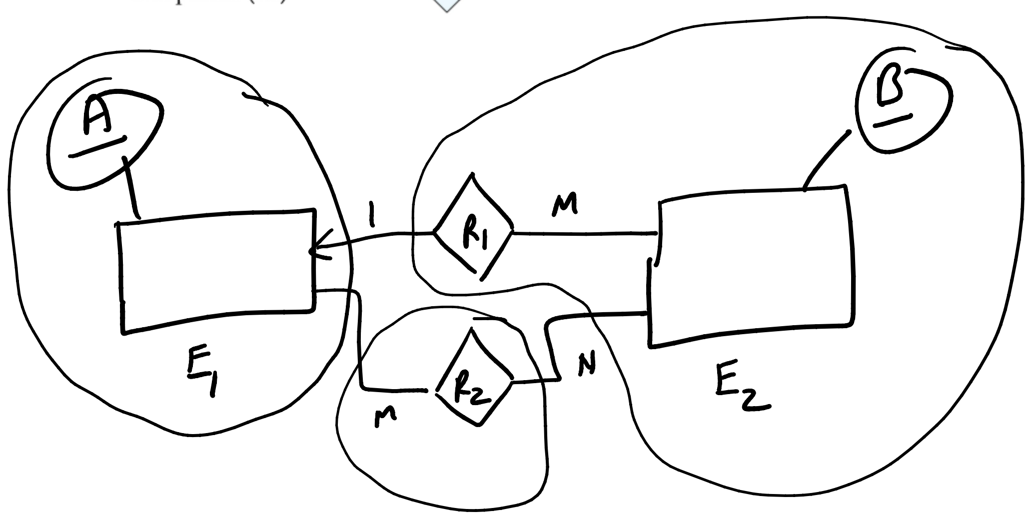

many-to-one
relationship




participation
in R: total (E1)
and partial (E2)




Q.

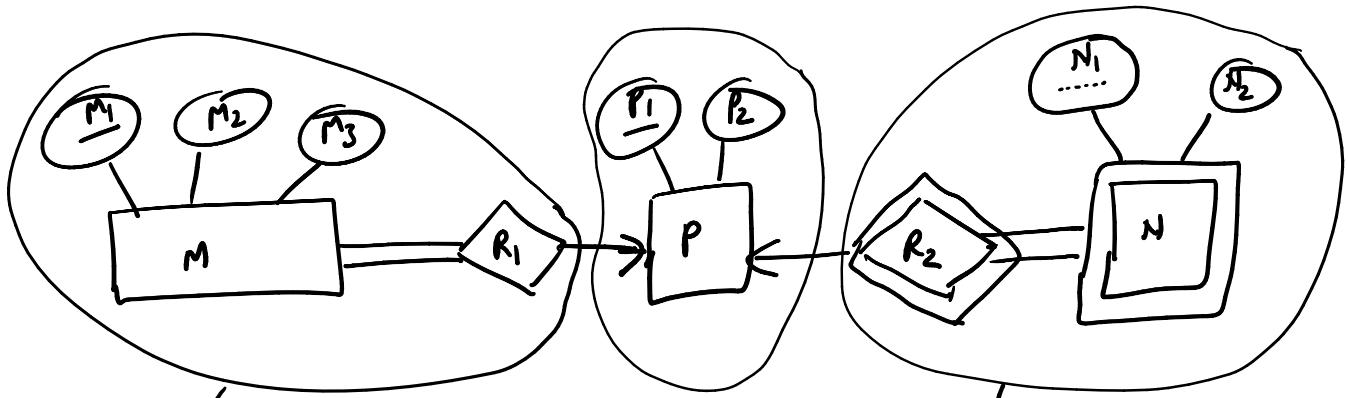



$E_1(\underline{A})$

$R_1E_2(\underline{B}, A)$

$R_2(\underline{AB})$

Q.

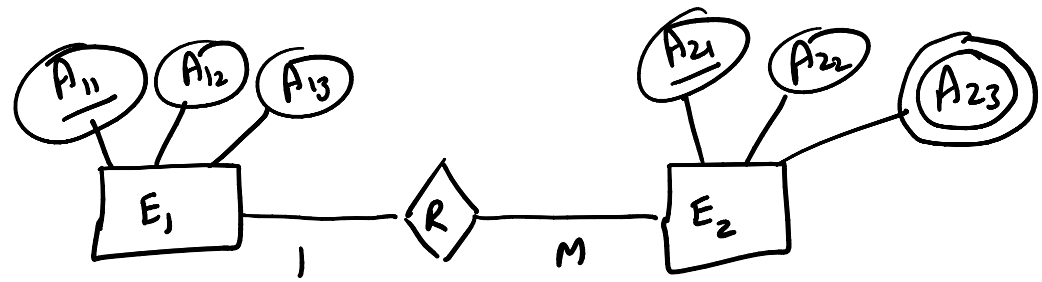



($\underline{M_1}$, $M_2$, $M_3$, $P_1$)

($\underline{P_1}$, $P_2$)

($\underline{P_1\ N_1}$, $N_2$)

Q.




$E_1(\underline{A_{11}}, A_{12}, A_{13})$

$E_2(\underline{A_{21}}, A_{22}, A_{23})$

$R(A_{11}, \underline{A_{21}})$

$E_1(\underline{A_{11}}, A_{12}, A_{13})$

$E_2R(\underline{A_{21}}, A_{22}, A_{11}, A_{23})$

$T_1$ $(\underline{A_{21}, A_{23}})$

$T_2$ $(\underline{A_{21}}, A_{22}, A_{11})$